

The 2025 AI Index Report | Stanford HAI

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The 2025 AI Index Report

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Top Takeaways

Measuring trends in Intelligence

The AI Index report tracks, collates, distills, and visualizes data related to artificial intelligence (AI). Our mission is to provide unbiased, rigorously vetted, broadly sourced data in order for policymakers, researchers, executives, journalists, and the general public to develop a more thorough and nuanced understanding of the complex field of AI.

[See Also: Report Revisions](#)

[Access the Public Data](#)

Policy Highlights

Policymakers use the AI Index to inform their understanding and decisions about AI. We curated a summary of highlights from the AI Index Report 2025 that are particularly relevant to policymakers and other policy audiences.

[Download the Policy Highlights](#)

Past Reports

[The 2026 AI Index Report](#)

[The 2024 AI Index Report](#)

Welcome to the seventh edition of the AI Index report. The 2024 Index is our most comprehensive to date and arrives at an important moment when AI ’ s influence on society has never been more pronounced.

[The 2023 AI Index Report](#)

The AI Index is an independent initiative at the Stanford Institute for Human-Centered Artificial Intelligence (HAI), led by the AI Index Steering Committee, an interdisciplinary group of experts from across academia and industry.

[The 2022 AI Index Report](#)

The 2021 AI Index Report

This year we significantly expanded the amount of data available in the report, worked with a broader set of external organizations to calibrate our data, and deepened our connections with Stanford HAI.

The 2019 AI Index Report

The AI Index Report tracks, collates, distills, and visualizes data relating to artificial intelligence.

Its mission is to provide unbiased, rigorous, and comprehensive data for policymakers, researchers, journalists, executives, and the general public to develop a deeper understanding of the complex field of AI.

The 2018 AI Index Report

Artificial Intelligence has leapt to the forefront of global discourse, garnering increased attention from practitioners, industry leaders, policymakers, and the general public. The diversity of opinions and debates gathered from news articles this year illustrates just how broadly AI is being investigated, studied, and applied. However, the field of AI is still evolving rapidly and even experts have a hard time understanding and tracking progress across the field.

The 2017 AI Index Report

Chapter Lineup

Chapter 1: Research and Development

This chapter explores trends in AI research and development, beginning with an analysis of AI publications, patents, and notable AI systems.

Chapter 2: Technical Performance

The Technical Performance section of this year ' s AI Index provides a comprehensive overview of AI advancements in 2024.

Chapter 3: Responsible AI

Artificial intelligence is now deeply integrated into nearly every aspect of our lives. It is reshaping sectors like education, finance, and healthcare, where algorithm-driven insights guide critical decisions.

Chapter 4: Economy

Global private AI investment hits record high...

Chapter 5: Science and Medicine

This chapter explores key trends in AI-driven science and medicine, reflecting the technology ' s growing impact in these fields.

Chapter 6: Policy and Governance

AI ' s advancing capabilities have captured policymakers ' attention, leading to an increase in AI-related policies worldwide.

Chapter 7: Education

AI has entered the public consciousness through generative AI ' s impact on work...

Chapter 8: Public Opinion

As AI continues to permeate broad swaths of society, it is becoming increasingly important to understand public sentiment around the technology.

AI ' s influence on society has never been more pronounced.

At Stanford HAI, we believe AI is poised to be the most transformative technology of the 21st century. But its benefits won ' t be evenly distributed unless we guide its development thoughtfully. The AI Index offers one of the most comprehensive, data-driven views of artificial intelligence. Recognized as a trusted resource by global media, governments, and leading companies, the AI Index equips policymakers, business leaders, and the public with rigorous, objective insights into AI ' s technical progress, economic influence, and societal impact.

New this Year: The Official Chinese Version of the 2025 AI Index Report

[Read the translation](#)

1. AI performance on demanding benchmarks continues to improve.

In 2023, researchers introduced new benchmarks—MMMU, GPQA, and SWE-bench—to test the limits of advanced AI systems. Just a year later, performance sharply increased: scores rose by 18.8, 48.9, and 67.3 percentage points on MMMU, GPQA, and SWE-bench, respectively. Beyond benchmarks, AI systems made major strides in generating high-quality video, and in some settings, language model agents even outperformed humans in programming tasks with limited time budgets.

2. AI is increasingly embedded in everyday life.

From healthcare to transportation, AI is rapidly moving from the lab to daily life. In 2023, the FDA approved 223 AI-enabled medical devices, up from just six in 2015. On the roads, self-driving cars are no longer experimental: Waymo, one of the largest U.S. operators, provides over 150,000 autonomous rides each week, while Baidu's affordable Apollo Go robotaxi fleet now serves numerous cities across China.

3. Business is all in on AI, fueling record investment and usage, as research continues to show strong productivity impacts.

In 2024, U.S. private AI investment grew to \$109.1 billion—nearly 12 times China's \$9.3 billion and 24 times the U.K.'s \$4.5 billion. Generative AI saw particularly strong momentum, attracting \$33.9 billion globally in private investment—an 18.7% increase from 2023. AI business usage is also accelerating: 78% of organizations reported using AI in 2024, up from 55% the year before. Meanwhile, a growing body of research confirms that AI boosts productivity and, in most cases, helps narrow skill gaps across the workforce.

4. The U.S. still leads in producing top AI models—but China is closing the performance gap.

In 2024, U.S.-based institutions produced 40 notable AI models, significantly outpacing China's 15 and Europe's three. While the U.S. maintains its lead in quantity, Chinese models have rapidly closed the quality gap: performance differences on major benchmarks such as MMLU and HumanEval shrank from double digits in 2023 to near parity in 2024. Meanwhile, China continues to lead in AI publications and patents. At the same time, model development is increasingly global, with notable launches from regions such as the Middle East, Latin America, and Southeast Asia.

5. The responsible AI ecosystem evolves—unevenly.

AI-related incidents are rising sharply, yet standardized RAI evaluations remain rare among major industrial model developers. However, new benchmarks like HELM Safety, AIR-Bench, and FACTS offer promising tools for assessing factuality and safety. Among companies, a gap persists between recognizing RAI risks and taking meaningful action. In contrast, governments are showing increased urgency: In 2024, global cooperation on AI governance intensified, with organizations including the OECD, EU, U.N., and African Union releasing frameworks focused on transparency, trustworthiness, and other core responsible AI principles.

6. Global AI optimism is rising—but deep regional divides remain.

In countries like China (83%), Indonesia (80%), and Thailand (77%), strong majorities see AI products and services as more beneficial than harmful. In contrast, optimism remains far lower in places like Canada (40%), the United States (39%), and the Netherlands (36%). Still, sentiment is shifting: since 2022, optimism has grown significantly in several previously skeptical countries—including Germany (+10%), France (+10%), Canada (+8%), Great Britain (+8%), and the United States (+4%).

7. AI becomes more efficient, affordable and accessible.

Driven by increasingly capable small models, the inference cost for a system performing at the level of GPT-3.5 dropped over 280-fold between November 2022 and October 2024. At the hardware level, costs have declined by 30% annually, while energy efficiency has improved by 40% each year. Open-weight models are also closing the gap with closed models, reducing the performance difference from 8% to just 1.7% on some benchmarks in a single year. Together, these trends are rapidly lowering the barriers to advanced AI.

8. Governments are stepping up on AI—with regulation and investment.

In 2024, U.S. federal agencies introduced 59 AI-related regulations—more than double the number in 2023—and issued by twice as many agencies. Globally, legislative mentions of AI rose 21.3% across 75 countries since 2023, marking a ninefold increase since 2016. Alongside growing attention, governments are investing at scale: Canada pledged \$2.4 billion, China launched a \$47.5 billion semiconductor fund, France committed €109 billion, India pledged \$1.25 billion, and Saudi Arabia's Project Transcendence represents a \$100 billion initiative.

9. AI and computer science education is expanding—but gaps in access and readiness persist.

Two-thirds of countries now offer or plan to offer K – 12 CS education—twice as many as in 2019—with Africa and Latin America making the most progress. In the U.S., the number of graduates with bachelor ’ s degrees in computing has increased 22% over the last 10 years. Yet access remains limited in many African countries due to basic infrastructure gaps like electricity. In the U.S., 81% of K – 12 CS teachers say AI should be part of foundational CS education, but less than half feel equipped to teach it.

10. Industry is racing ahead in AI—but the frontier is tightening.

Nearly 90% of notable AI models in 2024 came from industry, up from 60% in 2023, while academia remains the top source of highly cited research. Model scale continues to grow rapidly—training compute doubles every five months, datasets every eight, and power use annually. Yet performance gaps are shrinking: the score difference between the top and 10th-ranked models fell from 11.9% to 5.4% in a year, and the top two are now separated by just 0.7%. The frontier is increasingly competitive—and increasingly crowded.

11. AI earns top honors for its impact on science.

AI ’ s growing importance is reflected in major scientific awards: two Nobel Prizes recognized work that led to deep learning (physics), and to its application to protein folding (chemistry), while the Turing Award honored groundbreaking contributions to reinforcement learning.

12. Complex reasoning remains a challenge.

AI models excel at tasks like International Mathematical Olympiad problems but still struggle with complex reasoning benchmarks like PlanBench. They often fail to reliably solve logic tasks even when provably correct solutions exist, limiting their effectiveness in high-stakes settings where precision is critical.